

Answer **all** the questions and **marks** are assigned against each question

Solve the numerical first on rough paper then write the steps and solution neatly here

1. Consider a single longwall panel in a seam mined full height of 5 m at a depth of 400 m. The panel has a face width of 300 m; panel length will eventually reach 3,000 m. The seam is flat. Mining occurs at a rate of 8,000 tons per shift (1 ton = 2,000 lbs), two production and one maintenance shift per day, 5 days per week. Assume UK conditions, then determine for a single panel:
 - a) maximum subsidence expected
 - b) critical area
 - c) subsidence trough width
 - d) maximum tensile strain
 - e) maximum compressive strain
 - f) severity of damage (25 m structure width)
 - g) compressive strain at the center of the rib-side subsidence profile
 - h) maximum subsidence when the panel has advanced only 200 m
 - i) time required to mine the panel
 - j) maximum subsidence after mining four panels side by side and
 - k) maximum subsidence after mining a single panel at 400 m and an identical panel 420 m depth directly below.
 - l) Assuming Utah conditions with a subsidence factor of 0.65 and an angle of draw of 28°, find the maximum subsidence and trough width for a single panel

(1 + 1 + 1 + 1 + 1 + 1 + 1 + 1 + 1 + 1 + 2 + 2 + 2 = 16 marks)

- a) maximum subsidence expected

$$\frac{W}{H} = \frac{300}{400} = 0.75$$

$$\frac{S_{max}}{h} = 0.9 \left\{ 1 - e^{[-2.47(\frac{W}{H})]^2} \right\} = 0.9 \left\{ 1 - e^{[-2.47(0.75)]^2} \right\} = 0.676$$

$$S_{max} = 0.676 * 5 = 3.38 \text{ m}$$

- b) critical area

$$W_c = 1.4 * H = 1.4 * 400 = 560 \text{ m}$$

$$A_c = 560 \text{ m} * 560 \text{ m} = 313600 \text{ m}^2$$

- c) subsidence trough width = $W + 2H \tan \delta = 300 + 2 * 400 * \tan 35 = 860 \text{ m}$

- d) maximum tensile strain

$$\frac{W}{H} = \frac{300}{400} = 0.75; E_+ = \frac{0.68+0.65}{2} = 0.665 \text{ and } E_- = \frac{0.86+0.70}{2} = 0.780$$

$$E_t = (E_+) \frac{S}{H} = 0.665 * \frac{3.38}{400} = 5.61925 * 10^{-3}$$

- e) maximum compressive strain

$$E_c = (E_-) \frac{S}{H} = 0.780 * \frac{3.38}{400} = 6.591 * 10^{-3}$$

- f) severity of damage (25 m structure width)

Severity of change: depends on structure length at $5.61925 * 10^{-3}$ tensile strain will be severe for 25 m long structure.

- g) compressive strain at the center of the rib-side subsidence profile

$$\frac{-e}{-E} = e^{[-4(\frac{W}{H}-0.5)^2]} = e^{[-4(0.75-0.5)^2]} = 0.78$$

$$e_c = 5.141 * 10^{-3}$$

- h) maximum subsidence when the panel has advanced only 200 m

$$\frac{S'}{S} = \left\{ 1 - e^{[-2.70(\frac{x}{H})]^2} \right\} = \left\{ 1 - e^{[-2.70(\frac{200}{400})]^2} \right\} = 0.49$$

$$S' = 0.49 * 3.40 = 1.66 \text{ m}$$

- i) time required to mine the panel

$$\text{Volume of coal in the panel} = 5 * 300 * 3000 = 4500000$$

W/H	E+	E-
0.10	0.25	2.40
0.20	0.52	2.25
0.30	0.70	2.00
0.40	0.77	1.73
0.50	0.80	1.35
0.60	0.75	1.06
0.70	0.68	0.86
0.80	0.65	0.70
0.90	0.65	0.60
1.00	0.65	0.55
1.10	0.65	0.54
1.20	0.65	0.53
1.30	0.65	0.52
1.40	0.65	0.51

Note

a Tension (+), compression (-).

Tonnage of coal in the panel = $4500000 \times 1.4 = 6300000$ tons of coal

No. of shifts required to mine the panel = $\frac{6300000}{8000} = 787.5$

No. of weeks required to mine the panel = $\frac{787.5}{5 \times 2} = 78.75$

j) maximum subsidence after mining four panels side by side and

$$\frac{W}{H} = \frac{4 \times 300}{400} = 3$$

$$\frac{S_{max}}{h} = 0.9 \left\{ 1 - e^{[-2.47(\frac{W}{H})]^2} \right\} = 0.9 \left\{ 1 - e^{[-2.47(3)]^2} \right\} = 0.89$$

$$S_{max} = 0.89 \times 5 = 4.45 \text{ m}$$

k) maximum subsidence after mining a single panel at 400 m and an identical panel 420 m depth directly below.

$$\left(\frac{S}{h}\right) = 0.9 \left\{ 1 - e^{[-2.47(\frac{W}{H})]^2} \right\} = 0.676$$

$$\therefore S = 5 \times 0.676 = 3.38 \text{ m (top)}$$

$$\left(\frac{S}{h}\right) = 0.9 \left\{ 1 - e^{[-2.47(\frac{W}{H})]^2} \right\} = 0.645$$

$$\therefore S = 5 \times 0.645 = 3.225 \text{ m (bottom)}$$

Superposing the maximums is permissible because the panels are aligned horizontally.

Thus, $S(\text{max}) = 3.38 + 3.224 = 6.604 \text{ m}$ under UK conditions.

l) Assuming Utah conditions with a subsidence factor of 0.65 and an angle of draw of 28° , find the maximum subsidence and trough width for a single panel

$$\text{Maximum Subsidence } S' = S \left(\frac{S_f'}{S_f} \right) = 3.38 \times \left(\frac{0.65}{0.90} \right) = 2.44 \text{ m}$$

$$\text{Trough Width} = W + 2H \tan \delta = 300 + 2 \times 400 \times \tan 28 = 725.36 \text{ m}$$

- 2. Three geophones are located along a tunnel at 10 m (G1), 310 m (G2), and 640 m (G3) from the tunnel entrance. Longitudinal wave velocity in the adjacent rock mass is 3,000 m/s. An event arrival is detected at G1 at 10:31:20.06 (hr, min, sec). Event arrivals are also detected at G2 at 10:31:20.04 and at G3 at 10:31:20.15. Determine if the event is locatable and if so, best estimates of location and time of the event.**

(3 marks)

$$\text{---}0\text{---}100\text{---}x_s, t_s\text{---}310\text{---}640\text{---}$$

The sketch shows the layout of geophones in the tunnel. In essence the geophones are in a line and an assumption is made that the event occurs between geophones G1 and G2. This assumption is guided by examination of travel time differences. In particular, the difference in time between G2 and G3 is just the time needed for the wave to travel between the two geophones.

$$(x_s - x_1) = v(t_1 - t_s) \text{ or } x_s - 10 = 3000(20.06 - t_s)$$

$$(x_2 - x_s) = v(t_2 - t_s) \text{ or } 310 - x_s = 3000(20.04 - t_s)$$

$$(x_3 - x_s) = v(t_3 - t_s) \text{ or } 740 - x_s = 3000(20.15 - t_s)$$

Solving for source location

$$310 - 2x_s + 10 = 3000(-20.06 + 20.04)$$

$$\therefore x_s = 190 \text{ m}$$

and for time of event

$$310 - 10 = 3000(20.06 + 20.04 - 2t_s)$$

$$\therefore t_s = 20$$

Thus, $x_s = 190 \text{ m}$ and $t_s = 10:31:20.00 \text{ (hr:min:sec)}$

- 3. A proposal for panel caving a massive ore zone using undercuts 250 ft by 600 ft (76.2 m $\times$ 183 m) in plan and 14 ft (5.4 m) high is made. If the bulking porosity is 0.075 and caving occurs directly over the undercut, how high above the undercut will the cave zone extend? Draw of ore removes solid material and creates additional void space. If caving extends to the surface, estimate the tons of ore drawn (12 ft³/ton, 0.34 m³/ton). If the draw rate is 50,000 tpd, what is the elapsed time from undercut completion until the draw appears at the surface?**

(5 marks)

Since there was no data related to depth was provided and missing in the question paper, therefore 5 marks will be awarded for finding the value of cave zone height (H) and other values in terms of depth.

Given: Let depth = D ft, panel cave 250 ft × 600 ft in plan, 14 ft high, bulking porosity B = 0.075, 12 ft³/ton & 50,000 tons per day.

(a) Cave zone height

$$B = \frac{V_v}{V} = \frac{V_v}{V_s + V_v} = \frac{h}{H + h} = \frac{14}{H + 14} = 0.075$$

$$H = 172.7 \text{ ft}$$

(b) Additional draw to reach surface

When caving extends to the surface

$$B = \frac{V'_v}{V'_v + V'_s}$$

$$V'_v + V'_s = (H' + H + h)A = V'$$

$$V'_v = B(H' + H + h)A = 0.075(D) * 250 * 600 = 11250 * D \text{ ft}^3$$

$$H' = 0.075D \text{ ft}$$

$$V'_v = V_v + \Delta V_s$$

$$V'_s = V_s - \Delta V_s$$

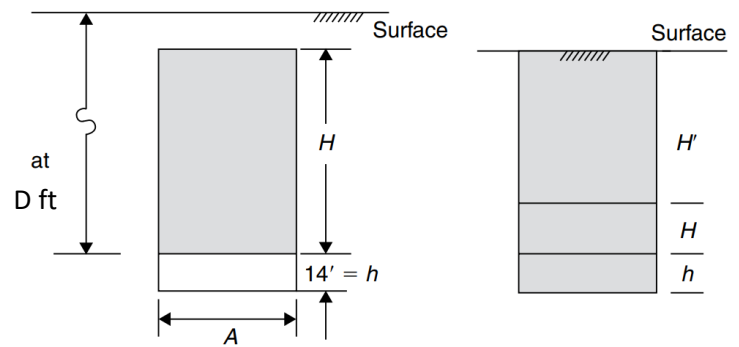
$$V'_v - V_v = \Delta V_v = \Delta V_s$$

$$\Delta V_v = 11250 * D - 14 * 250 * 600 \text{ ft}^3$$

$$\Delta V_v = \Delta V_s \text{ (new void comes from solid removed)}$$

$$\text{tons solid} = \frac{11250 * D - 14 * 250 * 600}{12} \text{ tons}$$

$$\text{time} = \frac{\text{tons}}{\text{tons/day}} = \frac{11250 * D - 14 * 250 * 600}{5 * 10^4} \text{ days}$$



4. Consider two vertical, circular shafts of the same diameter (7 m) in a pre-excitation stress field defined by $S_{xx} = 2.5, S_{yy} = 3.5, S_{zz} = 10.0, T_{yz} = 0.0, T_{zx} = 0.0, T_{xy} = -1.5$ where units are MPa, x = east, y = north, z = up. Determine the best orientation of the shaft pair and the stress concentrations in plan view at points A, B, and C defined in Figure. Assume the shafts are separated by 1.4 m. (5 marks)

According to rule of thumb, the row axis of multiple openings should be parallel to the major preexcavation compression. The direction of the preexcavation principal stress may be found by the solution of

$$\tan 2\alpha = \frac{\tau_{xy}}{\frac{\sigma_{xx} - \sigma_{yy}}{2}} = \frac{T_{xy}}{\frac{S_{xx} - S_{yy}}{2}} = \frac{-1.5}{\frac{2.5 - 3.5}{2}} = 3.0$$

$$\therefore 2\alpha = 71.6^\circ, 251.6^\circ$$

$$\therefore \alpha = 35.8^\circ, 125.8^\circ$$

The proper choice of the two solutions may be made by measuring from the axis of the largest preexcavation normal stress, in this example, the y-axis, and observing the sign convention that positive angles are measured counterclockwise, as shown in the sketch.

Numerical evaluation of stress concentrations at points A, B, and C shown in the sketch requires magnitudes of the preexcavation principal stresses. These are

$$\begin{aligned} \{S_1\} &= \left(\frac{S_{xx} + S_{yy}}{2} \right) \pm \sqrt{\left(\frac{S_{xx} - S_{yy}}{2} \right)^2 + (T_{xy})^2} \\ &= \left(\frac{2.5 + 3.5}{2} \right) \pm \sqrt{\left(\frac{2.5 - 3.5}{2} \right)^2 + (-1.5)^2} = 3 \pm 1.58 \end{aligned}$$

$$S_1 = 4.58 \text{ and } S_3 = 1.42$$

Accordingly, the ratio

$$M = \frac{1.42}{4.58} = 0.31$$

The pillar width to opening width ratio

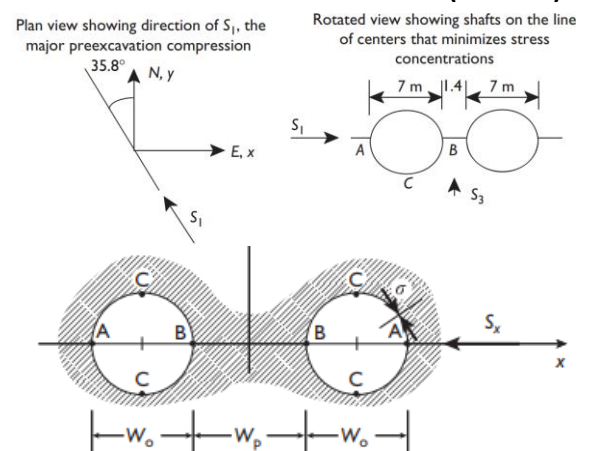


Table 3.6 Stress concentration about two identical circular holes

W_p/W_o	$S_x = 1.0, S_y = 0.0$			$S_x = 0.0, S_y = 1.0$		
	A	C	B	A	C	B
0.0		2.569		3.869		
0.1	-0.918	2.568	-0.651	3.414	-0.958	6.106
0.2	-0.905	2.580	-0.423	3.289	-0.924	4.423
0.3	-0.899	2.593	-0.351	3.231	-0.905	3.768
0.4	-0.896	2.608	-0.346	3.185	-0.892	3.443
0.5	-0.896	2.623	-0.377	3.151	-0.884	3.264
1.0	-0.908	2.703	-0.609	3.066	-0.881	3.020
2.0	-0.94	2.825	-0.838	3.020	-0.92	2.992
4.0	-0.927	2.927	-0.948	3.004	-0.964	2.997
7.0	-0.987	2.970	-0.981	3.001	-0.985	2.998
10.0	-0.993	2.984	-0.991	3.000	-0.992	3.000

$$\frac{W_p}{W_o} = \frac{1.47}{7.0} = 0.2$$

With S_1 and S_3 in the roles of S_{xx} and S_{yy} in Table, the stress concentrations are

$$K_A = -0.905 + 0.31(3.289) = 0.11$$

$$K_B = -0.423 + 0.31(4.423) = 0.95$$

$$K_C = 2.580 + 0.31(-0.924) = 2.29$$

There is no tension and the peak compressive stress concentration is 2.29. A single shaft would have a peak stress concentration of 2.69, about 14% higher in compression, and a tensile stress concentration of -0.07.

5. A circular concrete liner in a vertical shaft is subject to an unknown external pressure that cause the liner to fail in compression on the inside of the liner wall where traces of vertical shear cracks are observed. Concrete strength is 5,000 psi. The inside diameter of the liner is 18 ft and the thickness is 7.52 in. Determine the liner load at failure.

(3 marks)

At failure, the liner safety factor is 1.0; according to (3.20), this safety factor is

$$7.52 \text{ in} = \frac{7.52}{12} \text{ ft} = 0.63$$

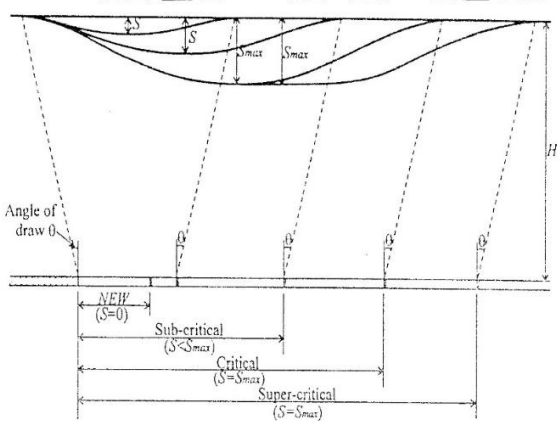
$$FS_c = C_o \left[\frac{1 - \left(\frac{a}{b}\right)^2}{2p_b} \right]$$

$$\therefore p_b = C_o \left[\frac{1 - \left(\frac{a}{b}\right)^2}{2FS_c} \right] = 5000 * \left[\frac{1 - \left(\frac{9}{9.63}\right)^2}{2FS_c} \right] = 315 \text{ psi}$$

The small thickness leads to a relatively small failure load.

A one-foot-thick liner would require a pressure of 475 psi, about 50% greater, to cause failure.

6. Differentiate between influence function versus profile function used to estimate/predict subsidence.	(1 mark) Influence function describes how subsidence at a point on the surface is influenced by a unit of extraction at depth. Profile function that models the shape of the subsidence trough (usually along a cross-section or profile line).
7. Discuss the rock mechanics rule with respect to separation distance between them and orientation to maintain safety for three shafts in a row.	(1 mark) Excavation should be separated (wall-to-wall) by at least one "diameter" to reduce interaction to an acceptable amount. Most favorable orientation of a row of identical excavation is with the axis parallel to the major principal preexcavation compression.
8. How Chinese mining industries have successfully extracted 35.5 m thick coal seam?	(1 mark) Longwall top-coal caving technology. This involves: Installation of 3 LTCC packages (two-pass mining: The lower portion (about 3–5 m) is cut using a shearer, while the remaining upper coal (top coal) is caved and collected behind the shield supports in two lifts. Specialized hydraulic supports: Designed to control massive roof pressure and allow efficient caving of thick seams. Advanced monitoring and automation: Ensures safety, improves recovery rates, and manages strata behavior effectively.
9. Define paste.	(1 mark) Potvin defined "paste as a mixture of material, which akin to toothpaste, comprising 75–85% solid by weight and 1 wt% to 12 wt% binder, containing at least 15 wt% solid fraction <20 µm, when poured into voids the backfill does not bleed water, does not settle in the pipeline, and has a slump <250 mm.
10. Discuss the significance of slump test with respect to paste backfilling.	(1 mark) Flowability of paste

	workability or consistency of paste yield stress of paste	
11. Differentiate between continuous and discontinuous subsidence.	(1 mark) Continuous subsidence is smooth, gradual, and widespread, often forming a trough over mined areas. Discontinuous subsidence is sudden and localized, causing abrupt ground movements like cracks or sinkholes.	
12. State the two major potential reasons responsible for failure of longwall technology in Indian coalfields.	(1 mark) No indigenous development. Mismatched imported Powered Roof Support with poor knowledge of strata behaviors.	
13. Differentiate NEW, sub-critical critical, supercritical panel widths using a suitable diagram.	(1 mark) 	

QUESTION PAPER ENDS HERE